An Introduction to Agent-Based Modeling

Unit 2: Building a Simple Model

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NetLogo

- Go through the tutorials on the NetLogo website
- Three Tabs
  - Interface
  - Info
  - Code

Model Settings

- World Size
- Wrapping

Running A Model

- Setup
- Go
- Speed Slider

Interface Elements

- Button
- Slider
- Chooser
- Switch
- Input
- Monitor
- Plot
- Output
- Note

Info Tab

- What is It?
- How It Works
- How to Use It
- Thing to Notice
- Things to Try
- Extending the Model
- NetLogo Features
- Related Models
- Credits and References

Turtle Relevant Commands

- create-turtles (crt)
- ask
- forward (fd), backward (bk)
- left (lt), right (rt)
- repeat
- color, size, xcor, ycor
- pen-down (pd), pen-up (pu)
- clear-all (ca)
- monitor
- die

Patches

- Inspector
- Patch Color is pcolor
- Turtles can directly access patches
- Relevant Commands
  - setxy, facexy
  - random-xcor (pxcor) and random-ycor (pycor)

Links

- Creating - create-link(s)-with / to /from
- Links have their own properties

Code Tab

- creating a procedure
  - to and end
- finding procedures
- indentation
- checking code

Saving and Documenting Your Code

- Save Often
- Save major changes with a new name
- Edit the Info Tab at the same time

Properties

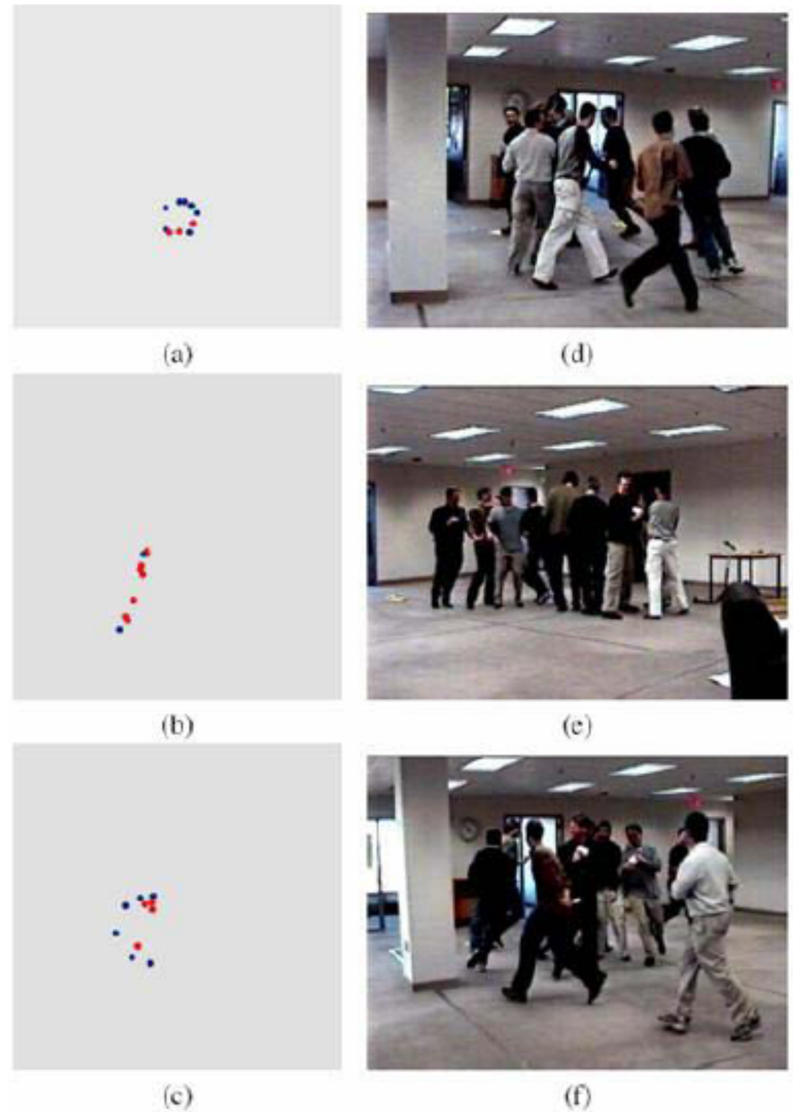
- global properties
  - globals
- turtle properties
  - turtles-own

Setup and Go

- setup and go are not required, but are NetLogo style
- Other commands:
  - tick
  - if, ifelse
  - repeat
  - while

Heroes and Cowards

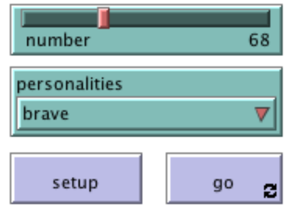
- Used by the Fratelli Group in the 1980s and 1990s and presented at a conference called Embracing Complexity in 1999
- Normally, to play the game you need a group of people, but we will explore it in simulation, several exist already
- Each person chooses an “enemy” and a “friend”
- Two stages to the game:
    - Cowards - place your friend between you and the enemy
    - Heroes - place yourself between your friend and the enemy

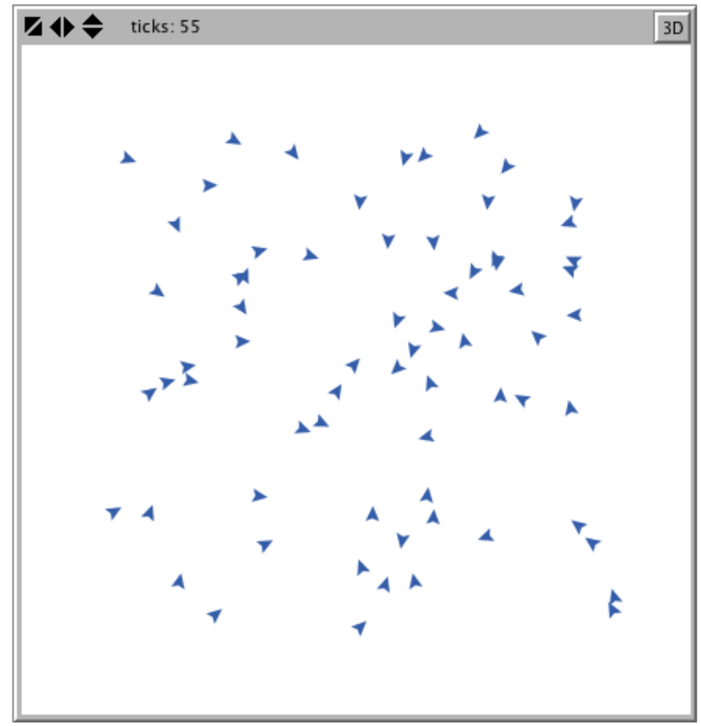


from Bonabeau et al., 2003

The *setup* procedure

```
to setup
    clear-all
    ask patches [ set pcolor white ]  ;; blank background
    create-turtles 100 [
        setxy random-xcor random-ycor
        ;; set the turtle personalities based on a chooser
        if (personalities = “cowards”) [ set color blue ]
        if (personalities = “heroes”) [ set color red ]
        ;; choose friend and enemy targets
        set friend one-of other turtles
        set enemy one-of other turtles
    ]
    reset-ticks
end
```





The *go* procedure

```
to go
  ask turtles [
    if (color = blue) [ act-cowardly ]
    if (color = red) [ act-bravely ]
  ]
end
```

act-bravely and *act-cowardly*

```
to act-bravely
   ;; move toward the midpoint of your friend and enemy
   facexy ([xcor] of friend + [xcor] of enemy) / 2
               ([ycor] of friend + [ycor] of enemy) / 2
   fd 0.1
end
to act-cowardly
   ;; move toward the midpoint of your friend and enemy
   facexy [xcor] of friend + ([xcor] of friend - [xcor] of enemy)
            [ycor] of friend + ([ycor] of friend - [ycor] of enemy)
   fd 0.1
end
```

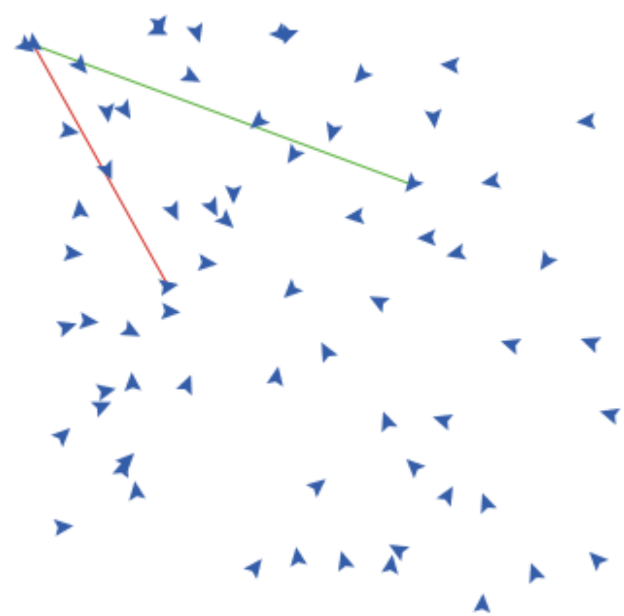
Extending the Model

- What about mixed types?

```
if (personalities = “mixed”) [ set color one-of [ red blue ] ]
```

Turtle Monitors and Links

- Inspect a turtle
- create-link-with friend [set color green]
- create-link-with enemy [set color red]



Documenting the Model

Initialize:

Create NUMBER turtles, where NUMBER is set by a slider in the interface
Each turtle moves to a random location on the screen
If the PERSONALITIES slider is set to “brave”, each turtle turns red
If the PERSONALITIES slider is set to “cowardly”, each turtle turns blue
If the PERSONALITIES slider is set to “mixed”, each turtle “flips a coin” and depending on the outcome, it turns red or blue
Each turtle picks one other turtle as a friend
Each turtle picks one other turtle as an enemy
The NetLogo clock is started

At each tick:

Each turtle asks itself “Am I red?” If yes, then I will act bravely by moving a step towards a location
between my friend and my enemy

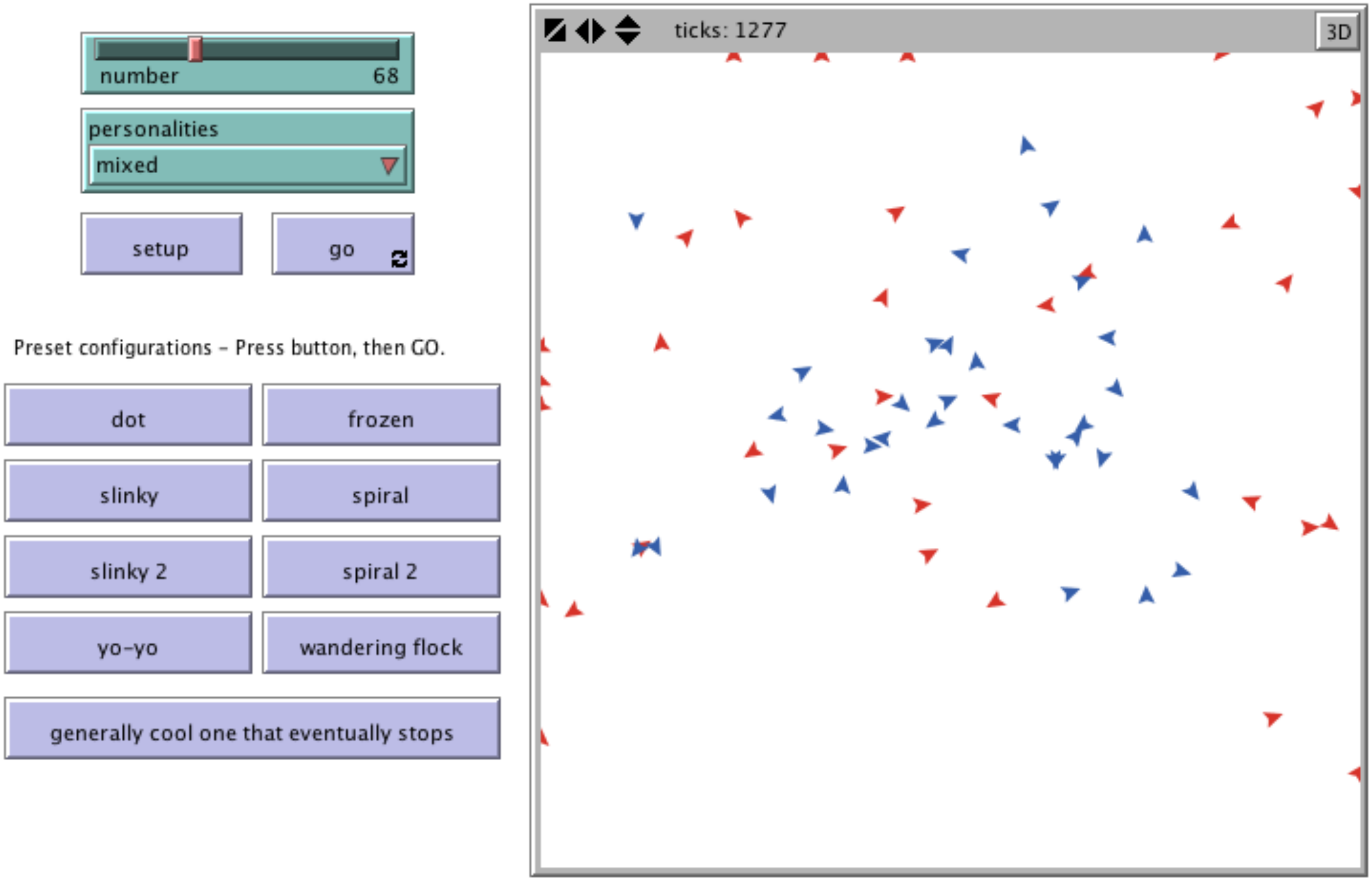
Each turtle asks itself “Am I blue?” If yes, then I will act cowardly by moving a step towards a location
that puts my friend between my enemy and me

Random Number Generators (RNG)

- RNGs - Create pseudo-random numbers
- Use a “seed” to create “random numbers” deterministically

```
random-seed 188
show random 100
show random 100
show random 100
random-seed 188
show random 100
show random 100
show random 100
```

Discovering Patterns



Unit 2 Wrap-Up

- Introduction to NetLogo
- Turtles, Patches, and Links
- The Code Tab
- Properties
- Heroes and Cowards
- Building the Model
- Extending the Model
- Documentation
- Random Numbers
- Guest Video by Marco Janssen at Arizona State University - Empirical ABM
- Unit 2 Test
- Slides
- Next Week: Unit 3 - Extending Models

Thank You